



Oscillatory Neural Network Based Edge Computing Architecture for Real-Time Obstacle Avoidance

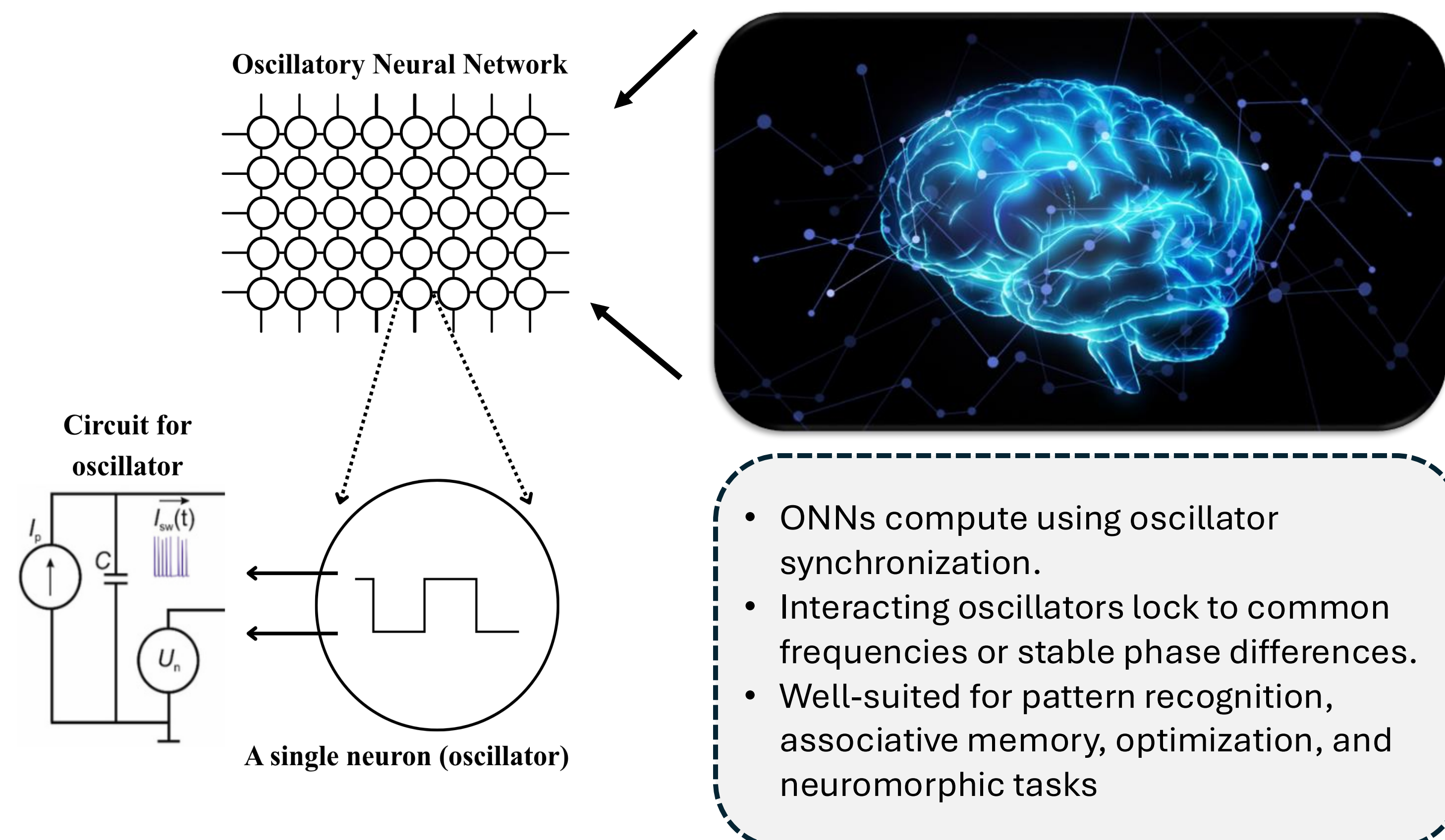
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What is ONN?

- Oscillatory Neural Networks (ONNs) are neural networks where the basic computing elements are oscillators instead of traditional neurons.
- Each oscillator generates a periodic signal, and information is encoded in its phase, frequency, or amplitude.



Why ONN for Edge Computing?

Edge computing performs data processing locally on embedded or sensor-level devices instead of the cloud.

CMOS Bottlenecks

- Power density
- Memory-logic divide
- Inefficient for NP-hard problems

NP-Hard Problems

- Graph optimization
- Travelling Salseman Problem
- Brute-force solution expensive

Neuromorphic Inspiration

- Brain uses oscillations
- Synchronization encodes decisions

Oscillatory Neural Networks

- Coupled oscillators → stable phases
- Optimization + pattern recognition
- Low-power parallel computing

ONN Working Process

Step 1: Hebbian Learning Rule

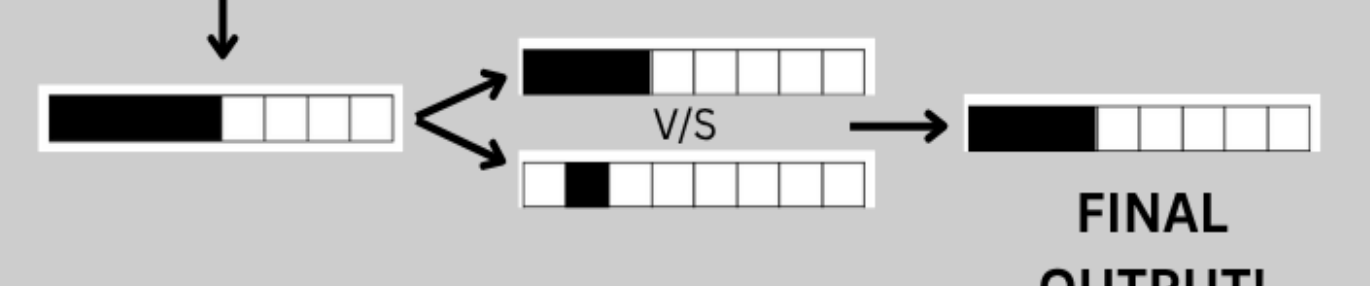
To generate a 8*8 weight matrix

	Pattern 1	Pattern 2
1	1	-1
2	1	1
3	1	-1
4	-1	-1
5	-1	-1
6	-1	-1
7	-1	-1
8	-1	-1

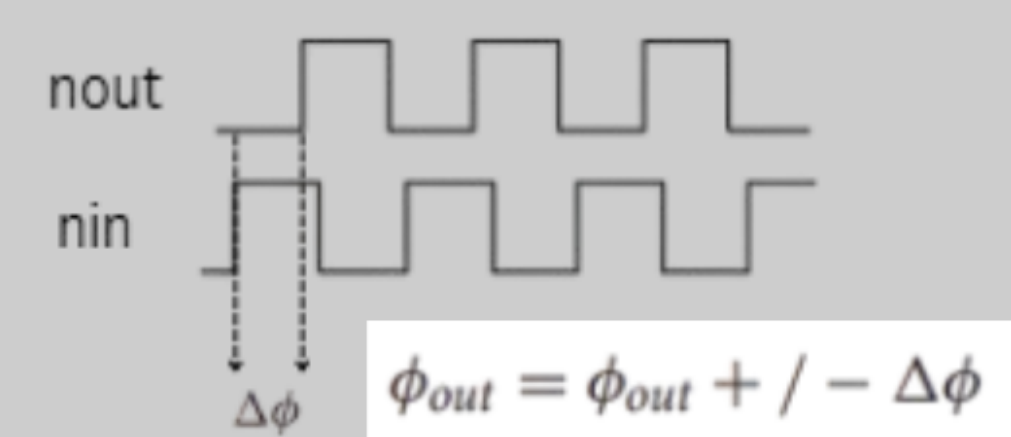
Using the phase as input to the multiplexer calculate the nout signal for each neuron

Step 5: Declaring the Final output pattern

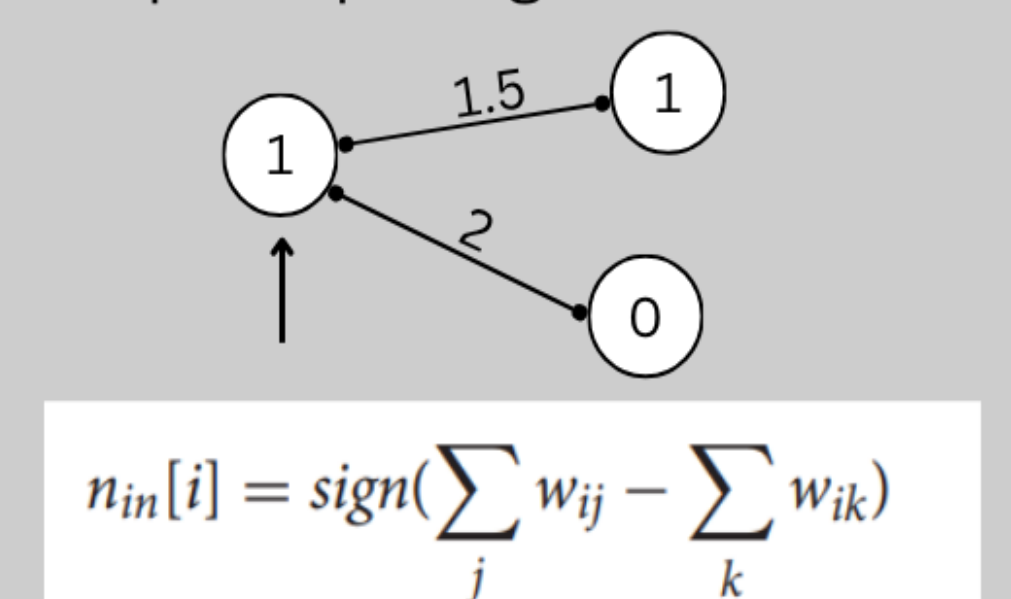
Suppose this is our final pattern after phase alignment



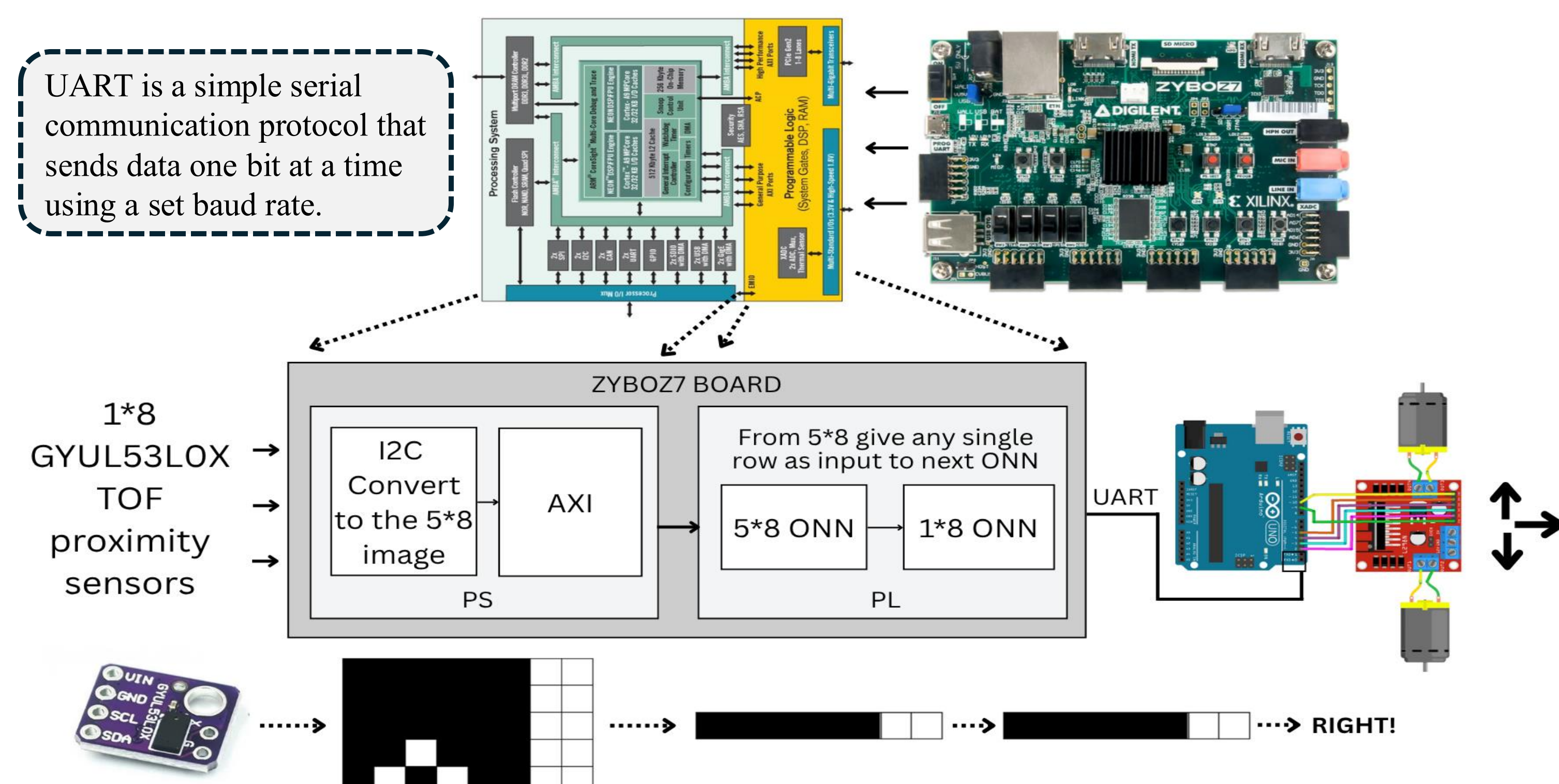
Step 4: Phase alignment



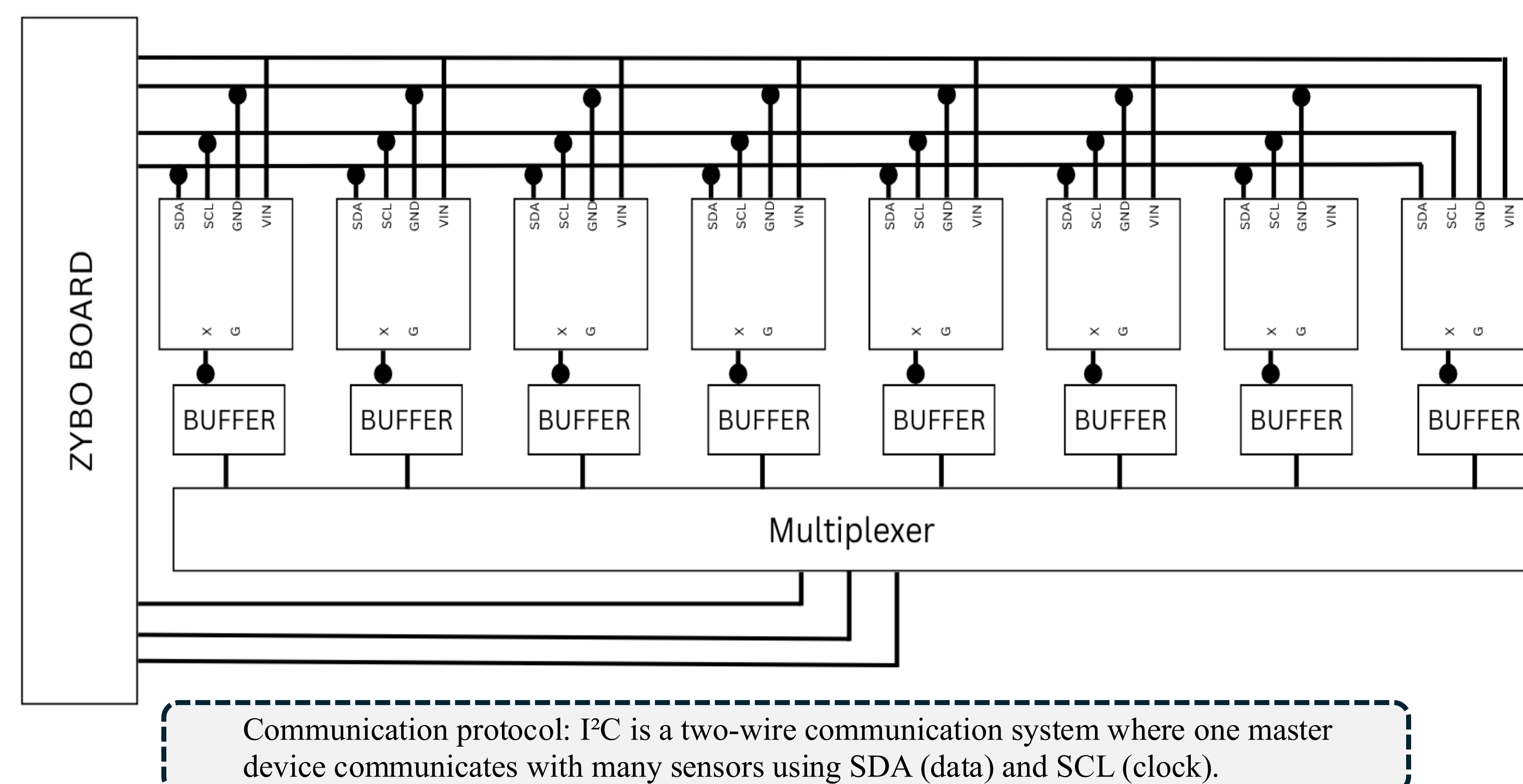
Step 3: Input signal calculation:



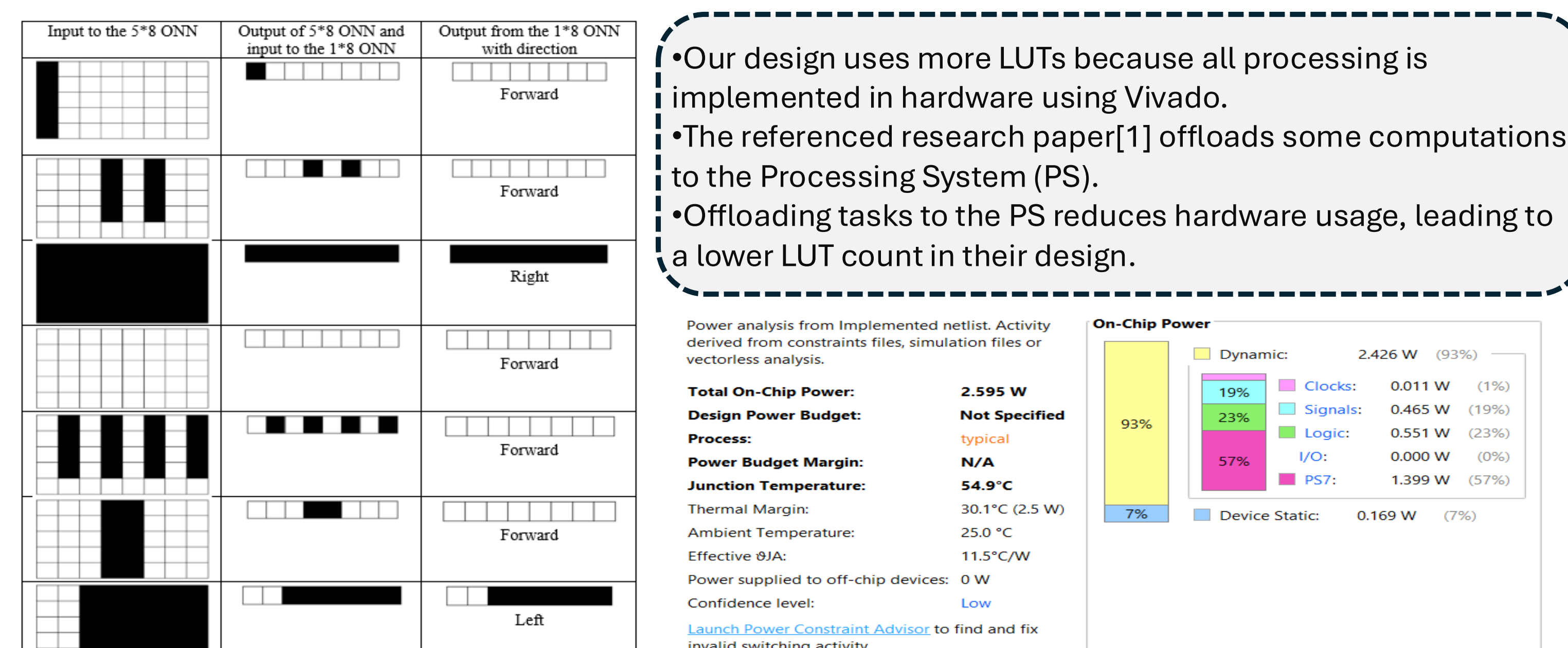
System Architecture



Hardware Implementation



Results and Analysis



Conclusion

- Developed a fast and low-power Oscillatory Neural Network (ONN) system for direction detection using oscillator synchronization instead of heavy computations.
- Integrated the GYUL53LOX ultra-proximity sensor to obtain accurate and stable short-range distance measurements.
- The buffer provided clean and reliable sensor inputs to the ONN, improving direction-estimation accuracy.
- Achieved smooth ONN-sensor integration, enabling stable and consistent real-time outputs.

References

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- [5] N. Liu, X. Zhang, Q. Ma, and B. Xu, "Oscillatory neural mechanisms of brain computation: A review," *Frontiers in Neuroscience*, vol. 17, 2023. [Online]